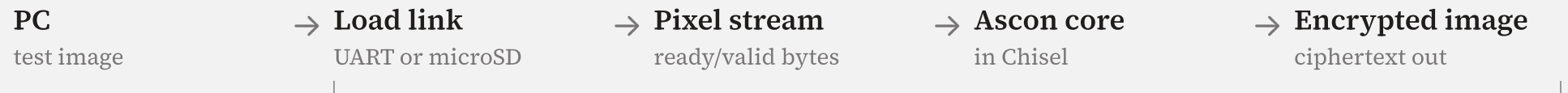


Encrypting images in hardware

- Image encryption accelerator on an FPGA
- Pixels stream in, get encrypted with Ascon, and stream out
- Real-time applications

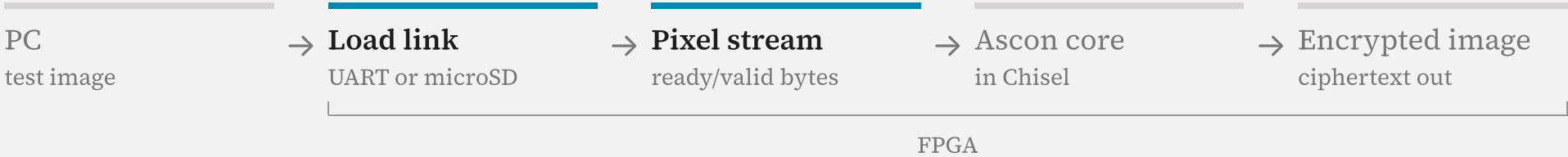


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From image to pixel stream

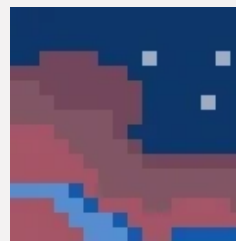
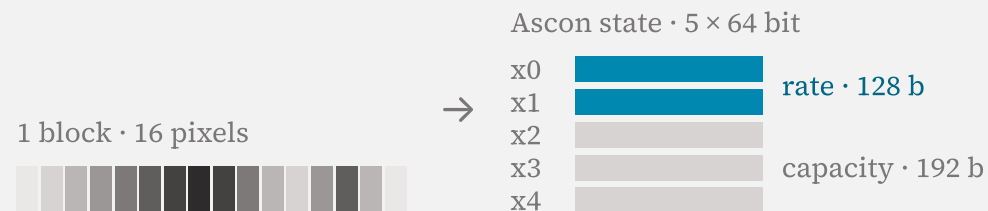
- Static 128×128 grayscale test image: 16 KB, fits in on-chip block RAM
- Loading link still open: USB-UART or microSD, both on the board
- The core sees a ready/valid byte stream, so it doesn't depend on the link

	USB-UART	MICROSD
On the board	FTDI USB-UART bridge	microSD slot
Host side	Serial script	Copy file to card
FPGA logic	UART receiver	SD controller
16 KB load	1.4 s at 115.2k baud	Read at start-up

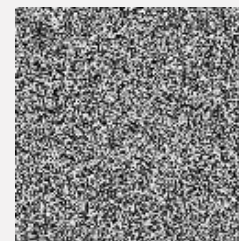


Ascon in hardware

- NIST lightweight standard: 320-bit state, rounds of only XOR, AND, NOT and rotations
- 16 pixels pack into one 16-byte block: the image is 1,024 blocks
- Design knob: rounds per clock cycle, trading area for throughput
- Behaves like a stream cipher: the output is noise with no visible outline



Original



Ascon: uniform noise

PC
test image

→ Load link
UART or microSD

→ Pixel stream
ready/valid bytes

→ **Ascon core**
in Chisel

→ Encrypted image
ciphertext out

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Why Chisel

- The design space is a set of parameters: datapath width and rounds per cycle
- Speed: trade security margin for throughput in time-constrained environments
- Color: RGB or grayscale datapath, for color-sensitive applications

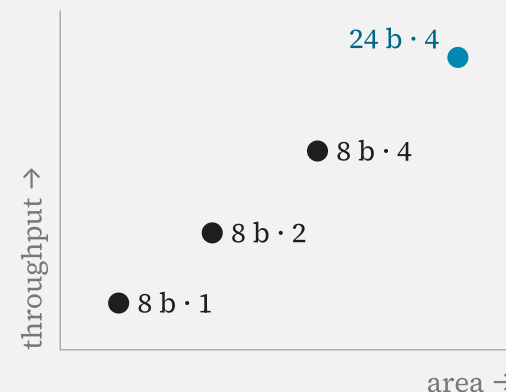
Chisel generator

width

8 b · 24 b

rounds / cycle

1 · 2 · 4



Illustrative: real points come from Artix-7 synthesis

PC
test image

→ Load link
UART or microSD

→ **Pixel stream**
ready/valid bytes

→ **Ascon core**
in Chisel

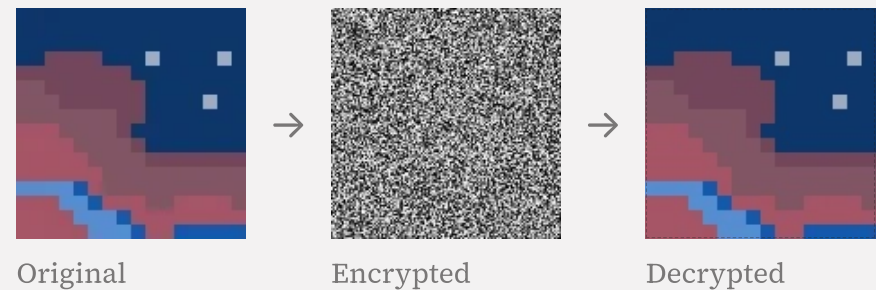
→ Encrypted image
ciphertext out

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MVP and roadmap

MVP, in simulation

- Ascon encryption core in Chisel, on the existing stream interface
- Stream the 128×128 grayscale image through it
- Ciphertext matches a software Ascon reference byte for byte
- Software decryption recovers the original image exactly



✓ Decrypted = original

Next steps

